Yiyang Feng

yiyang.feng@stonybrook.edu yiyangfeng.me Google Scholar Wind-2375-like yiyangfeng

Education

Stony Brook University — **Aug. 2025 – Present**
Ph.D. in Computer Science; Advisor: Prof. Jiawei Zhou — *Stony Brook, NY, USA*

École Polytechnique Fédérale de Lausanne (EPFL) — **Sep. 2022 – July 2025**
M.S. in Computer Science — *Lausanne, Switzerland*

Xi'an Jiaotong University (XJTU) — **Aug. 2018 – July 2022**
B.Eng. in Automation — *Xi'an, China*

Research Interests

- **Long-horizon reasoning**: how new, outdated, or wrong knowledge propagates through multi-step and causal reasoning
- **Experiential learning**: skills that transfer across tasks; computer-use agents that learn a user's workflows
- **Efficient agents**: context management and compression for long-horizon and multi-agent systems

Publications *(* equal contribution)*

First or co-first author

- **Break It Down, Pass It On: Cross-Task Skill Transfer in LLM Agents**
  Yiyang Feng, Biddut Sarker Bijoy, Niranjan Balasubramanian, Jiawei Zhou
  arXiv preprint, 2026 [paper] [code] [website]
- **Averaging Bias: Human Faithfulness Annotations are not Locally Faithful**
  Huajian Zhang*, **Yiyang Feng***, Jiawei Zhou
  arXiv preprint, 2026 [paper] [code]
- **Tracking the Limits of Knowledge Propagation: How LLMs Fail at Multi-Step Reasoning with Conflicting Knowledge**
  Yiyang Feng, Zeming Chen, Haotian Wu, Jiawei Zhou, Antoine Bosselut
  EACL 2026 (Main Conference), Virtual Oral [paper] [code] [website]
- **Unraveling Misinformation Propagation in LLM Reasoning**
  Yiyang Feng*, Yichen Wang*, Shaobo Cui, Boi Faltings, Mina Lee, Jiawei Zhou
  Findings of EMNLP 2025 [paper] [code] [website]
- **Unveiling the Art of Heading Design: A Harmonious Blend of Summarization, Neology, and Algorithm**
  Shaobo Cui*, **Yiyang Feng***, Yisong Mao, Yifan Hou, Boi Faltings
  Findings of ACL 2024; nominated as a best paper candidate by reviewers and meta-reviewers [paper]

Co-author

- **Self-Improvement of Large Language Models: A Technical Overview and Future Outlook**
  Haoyan Yang, Mario Xerri, Solha Park, Huajian Zhang, **Yiyang Feng**, Sai Akhil Kogilathota, Jiawei Zhou
  Transactions on Machine Learning Research (TMLR), 2026, Survey Certification [paper] [code]
- **FURINA: A Fully Customizable Role-Playing Benchmark via Scalable Multi-Agent Collaboration Pipeline**
  Haotian Wu*, Shufan Jiang*, Chios Chen, **Yiyang Feng**, Hehai Lin, Heqing Zou, Yao Shu, Chengwei Qin
  arXiv preprint, 2026 [paper]
- **Conditional Dichotomy Quantification via Geometric Embedding**
  Shaobo Cui, Wenqing Liu, **Yiyang Feng**, Jiawei Zhou, Boi Faltings
  ACL 2025 (Main Conference), Oral [paper]
- **Nuance Matters: Probing Epistemic Consistency in Causal Reasoning**
  Shaobo Cui, Junyou Li, Luca Mouchel, **Yiyang Feng**, Boi Faltings
  AAAI 2025 [paper]
- **Exploring Defeasibility in Causal Reasoning**
  Shaobo Cui, Lazar Milikic, **Yiyang Feng**, Mete Ismayilzada, Debjit Paul, Antoine Bosselut, Boi Faltings
  Findings of ACL 2024 [paper]

Research Experience

Skill Transfer and Context Management for LLM Agents **Aug. 2025 – Present**

Ph.D. research advised by Prof. Jiawei Zhou, with Prof. Niranjan Balasubramanian *Stony Brook University*

- Led a controlled study of how agents induce skills (task- vs. subtask-level, text vs. code) across 3 domains and 11 models; subtask-level skills raise agents above their no-memory baseline on average, while task-level skills mostly lower it
- Proposed a skill utility score (specificity $\times$ abstractness) that predicts transfer from skills and task descriptions alone
- In progress: context management for multi-agent systems built from small language models

Knowledge Propagation in Multi-Step Reasoning (TRACK) **Feb. 2025 – July 2025**

Master's thesis advised by Zeming Chen and Prof. Antoine Bosselut *NLP Lab, EPFL*

- Built TRACK, which probes each model's knowledge gaps and then injects conflicting facts (Wiki, Code, Math)
- Showed updated facts can make reasoning worse than none, increasingly so with more facts; EACL 2026 virtual oral

Misinformation Propagation in LLM Reasoning **Aug. 2024 – Aug. 2025**

Advised by Prof. Jiawei Zhou, with Prof. Mina Lee *Stony Brook University*

- Led the project; built a framework that injects user misinformation into math problems and traces it through each step
- Found models fix misinformation less than half the time even when told to, and early corrections help most

Causal Reasoning and Controllable Generation **Feb. 2023 – Dec. 2024**

Research assistant advised by Shaobo Cui and Prof. Boi Faltings *Artificial Intelligence Laboratory, EPFL*

- Co-led LOgogram, a heading-generation benchmark (co-first author); contributed to defeasible causal reasoning, causal epistemic consistency, and conditional dichotomy quantification (ACL 2025 oral)

Talks

Cross-task skill transfer in LLM agents. Lightning talk, North East AI Agents Day 2026, New York	**May 2026**
Tracking the limits of knowledge propagation. Virtual oral, EACL 2026	**Mar. 2026**
Unraveling misinformation propagation in LLM reasoning. Invited talk, Alan Turing Institute (online)	**Jan. 2026**
Unraveling misinformation propagation in LLM reasoning. Virtual poster, EMNLP 2025	**Nov. 2025**

Honors and Awards

Tinker Research Grant, Thinking Machines Lab	**2026**
Outstanding Reviewer, ACL Rolling Review (January 2026 cycle)	**2026**
Computer Science Chairman Fellowship, Stony Brook University	**2025, 2026**
NAIRR Pilot Compute Allocation, U.S. National Science Foundation (PI: Jiawei Zhou)	**2025**
Conference Travel Grant for Master's Students, EPFL	**2024**
Chiang Chen Overseas Fellowship (0.2‰), Chiang Chen Industrial Charity Foundation	**2022**
First Prize Scholarship (top 3%), Xi'an Jiaotong University	**2019**

Service and Teaching

Reviewer, ACL Rolling Review	**2025 – 2026**
Teaching Assistant, EPFL: Modern NLP (CS-552), Machine Learning (CS-433)	**2023 – 2025**

Skills

- Python, PyTorch, Hugging Face (Transformers, PEFT), FlashAttention, OpenAI and Together APIs, W&B, Slurm, LaTeX
- Chinese (native), English (fluent)